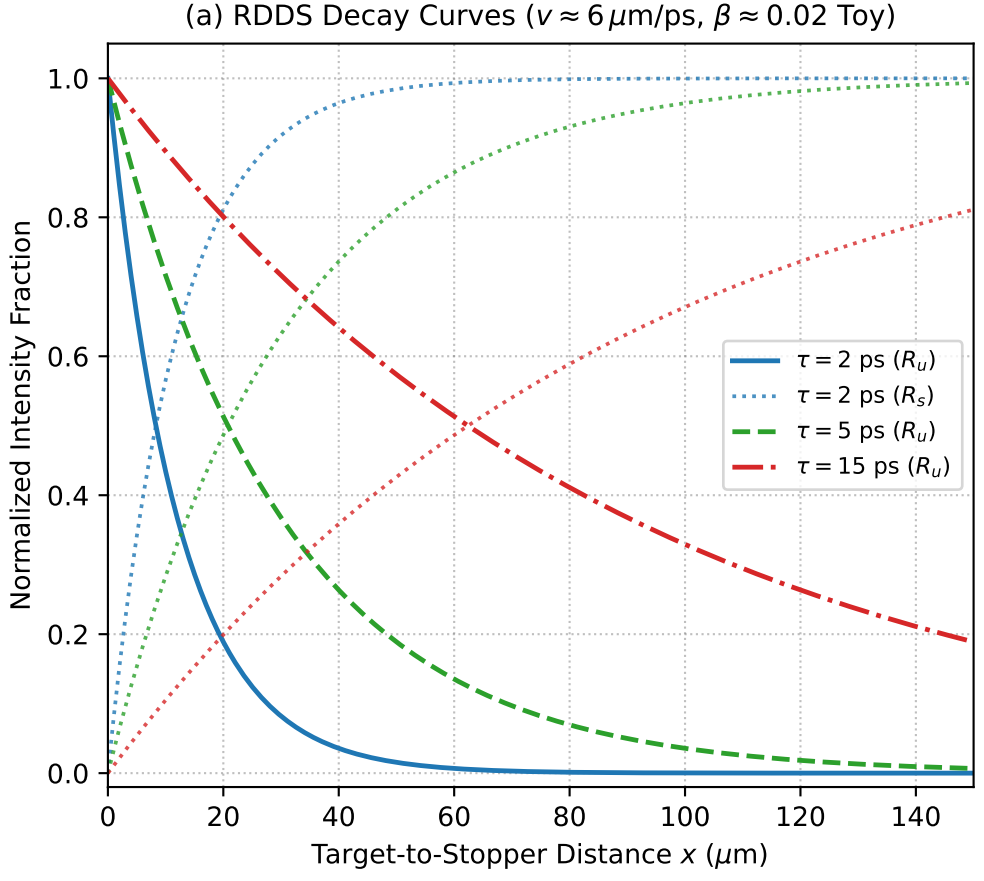
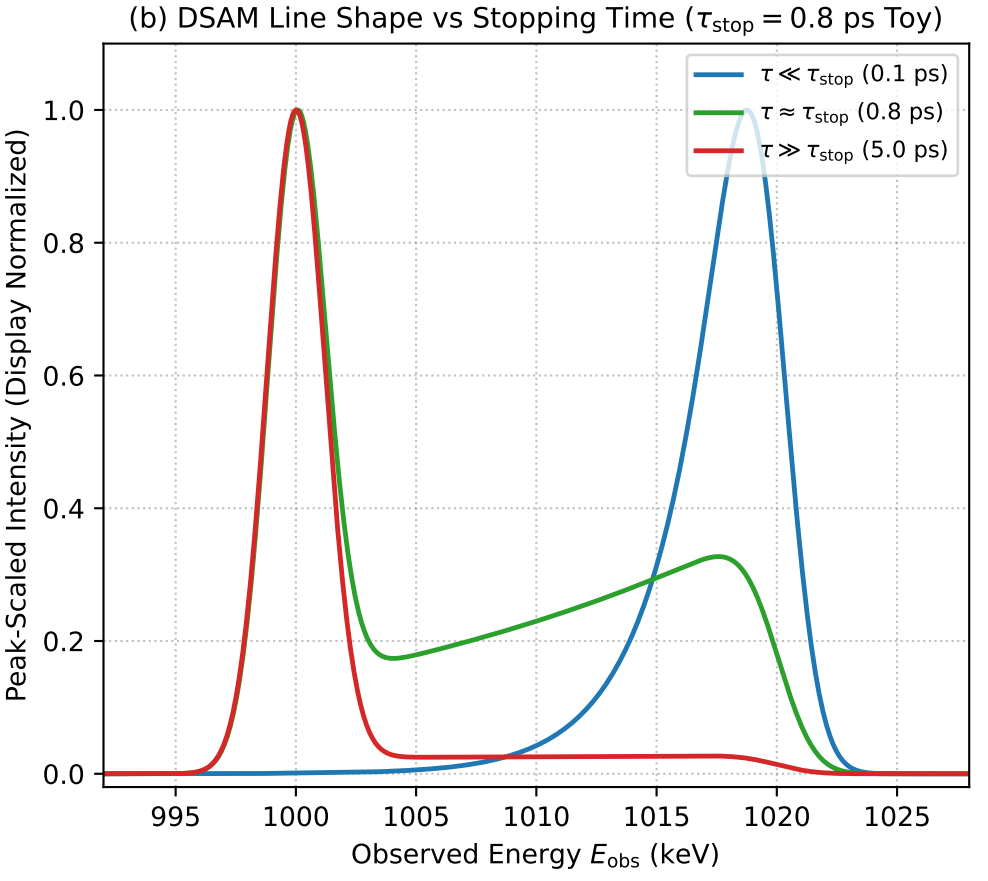

(a) RDDS Decay Curves ($v \approx 6\,\mu$m/ps, $\beta \approx 0.02$ Toy)
Normalized Intensity Fraction
Target-to-Stopper Distance x (μm)
$\tau = 2$ ps (R_u)
$\tau = 2$ ps (R_s)
$\tau = 5$ ps (R_u)
$\tau = 15$ ps (R_u)
0.0
0.2
0.4
0.6
0.8
1.0
0
20
40
60
80
100
120
140


(b) DSAM Line Shape vs Stopping Time ($\tau_{\mathrm{stop}} = 0.8$ ps Toy)
Peak-Scaled Intensity (Display Normalized)
Observed Energy E_{obs} (keV)
$\tau \ll \tau_{\mathrm{stop}}$ (0.1 ps)
$\tau \approx \tau_{\mathrm{stop}}$ (0.8 ps)
$\tau \gg \tau_{\mathrm{stop}}$ (5.0 ps)
0.0
0.2
0.4
0.6
0.8
1.0
995
1000
1005
1010
1015
1020
1025